

AKSHAI NARAYANAPPA

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Professional Summary

Data Engineer with 4+ years of experience building and maintaining scalable data pipelines on Google Cloud Platform using BigQuery, Cloud Spanner, Dataflow, Python, and Airflow. Hands-on experience in authentication and billing data processing, network experience data processing, REST API development for graph-based use cases, and building AI agent-based solutions. Skilled in SQL optimisation, data modelling, data validation, and end-to-end ETL development in a production environment.

TECHNICAL SKILLS

- **Programming & Query Languages:** Python, SQL (BigQuery, Cloud Spanner, MySQL, Hive SQL)
- **Cloud & Data Platforms:** Google Cloud Platform (GCP), BigQuery, Cloud Spanner, Dataflow, Teradata
- **ETL & Workflow Orchestration:** Airflow, Dataflow, ETL Pipelines
- **APIs & AI:** REST API development, AI Agents / LLM-based solutions
- **Tools:** Jupyter Notebook, VS Code, GitLab, Jenkins, Jira

WORK EXPERIENCE

Prodapt Solutions | Bangalore, India

Software Engineer (Oct 2023 – Present)

- Designing and developing AI agent-based solutions on the DEAH platform, leveraging LLMs and agentic workflows to automate data operations and analytical tasks.
- Designed and executed Dataflow-based ETL pipelines to load high-volume network experience data from BigQuery into Cloud Spanner, supporting graph-based analytical use cases.
- Owned end-to-end ETL pipelines on Google Cloud Platform, enabling reliable ingestion, transformation, and loading of production datasets into BigQuery, supporting daily analytical workloads.
- Implemented data validation and reconciliation frameworks, ensuring high data accuracy across pipelines and reducing operational downtime.

Associate Software Engineer (May 2022 – Oct 2023)

- Led the migration of 200+ Hive SQL queries to BigQuery, improving query performance while reducing processing costs.
- Designed and managed Airflow DAGs to automate ETL workflows, reducing task failures by 25% and improving pipeline reliability.

PROJECTS

DEAH – AI Agent Development

- Working on the DEAH platform, designing and developing AI agents for data engineering and analytics workflows.
- Built agent-based components that orchestrate multi-step tasks across data sources, leveraging LLMs to interpret natural-language requests and trigger downstream pipelines and queries.
- Integrated agents with existing GCP data infrastructure (BigQuery, Cloud Spanner) to enable automated data retrieval, transformation, and reporting.
- Focused on prompt design, tool integration, and reliability/guardrails to ensure consistent and accurate agent behavior in production.

Network Experience Data Processing (BigQuery → Cloud Spanner)

- Designed and executed ETL to load high-volume network experience data from BigQuery to Cloud Spanner using Dataflow, enabling low-latency access for graph-based use cases.
- Designed an optimised Cloud Spanner data model (schema, primary keys, interleaving, and indexes) to support scalability and high query performance for graph workloads.
- Part of REST API development for graph-based use cases, exposing Spanner-backed data to downstream applications and analytical consumers.
- Tuned Dataflow pipelines for throughput, cost, and reliability, handling large-scale daily ingestion from BigQuery into Spanner.

Modernisation of Hive SQL to BigQuery & DAG Optimisation

- Migrated 200+ Hive SQL queries to BigQuery, improving query performance by 40% and reducing compute cost by 30%.
- Designed and optimised 30+ Airflow DAGs, strengthening ETL orchestration, dependency management, and workflow reliability.
- Developed 100+ account-specific SQL scripts, enabling accurate, customised data transformations across multiple clients.
- Reduced manual operational effort by 50% through workflow automation and improved scheduling logic.

Data Ingestion, Replication & Email Automation

- Built end-to-end data ingestion pipelines from Oracle to BigQuery, supporting reliable batch data processing on GCP.
- Designed cross-platform data replication workflows between GCP and Teradata, ensuring consistency across analytical systems.
- Implemented automated email-based data reconciliation for GCP vs Teradata and Splunk, improving monitoring and issue detection.
- Optimised replication workflows, reducing data synchronisation latency by 30%.

Billing Data Processing & Business Logic Automation

- Owned and maintained production-grade ETL pipelines for billing data processing using Airflow, Python, and BigQuery on GCP.
- Implemented complex business logic for billing metrics, including current bill, previous bill, and bill difference calculations.
- Performed data validation and reconciliation to ensure accurate, audit-ready billing outputs in production environments.
- Optimised SQL queries to improve the performance, reliability, and scalability of billing workloads.

LEADERSHIP & ADMINISTRATIVE EXPERIENCE

- Mentored new data engineers on BigQuery, Cloud Spanner, Dataflow, Airflow, and ETL best practices, and led cross-team collaborations with data analysts, business teams, and cloud engineers.

CERTIFICATION & AWARDS

- **Certification:** GCP Associate Cloud Engineer, GCP Professional Machine Learning Engineer.
- **Awards:** "Star of the Month" for outstanding contributions.

EDUCATION

GITAM UNIVERSITY Bangalore, India

B.Tech in Computer Science & Engineering (CGPA: 8.02) **August 2018 – May 2022**